

# DUA E ZAINAB

AI & Software Development Intern Candidate | AI Student @ COMSATS University

📍 Fateh Jang, Punjab | 📞 0325-3161707 | ✉️ duaezainabimmam@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## OBJECTIVE

Motivated AI & Computer Science student seeking an internship in software development, data analysis, or AI/ML. Proven ability to build production-grade systems — including encrypted medical vaults and full-stack dashboards — while maintaining a strong academic record. Ready to contribute immediately and grow fast within a dynamic team.

## EDUCATION

### BS Artificial Intelligence

2023 – 2027 (Expected)

- Relevant coursework: Data Structures, OOP, Database Systems, AI Fundamentals, Machine learning, Deep learning, Java, Networking, web / Mobile dev, linear algebra, design and analysis of algo.

## PROJECTS & EXPERIENCE

### AlphaForge — Quant AI Trading Platform *Python · FastAPI · PyTorch · Reinforcement Learning · React*

- Built a full-stack algorithmic trading SaaS platform featuring DQN and PPO reinforcement learning agents trained on real financial market data. Implemented a custom Gym-compatible trading environment with portfolio tracking, slippage simulation, and risk-adjusted reward functions. Developed a backtesting engine computing Sharpe ratio, max drawdown, Calmar ratio, and win rate across multi-asset portfolios (BTC, ETH, AAPL, SPY). Includes a real-time training dashboard with SSE streaming, agent leaderboard, trade signal visualization, and a FastAPI backend with 20+ endpoints backed by SQLite.

### VisionOps AI – Real-Time MLOps Object Detection Platform

- Built a full-stack MLOps system for real-time object detection using YOLOv8 and ONNX Runtime Web. Integrated MLflow for experiment tracking, FastAPI for backend inference, and CI/CD pipelines using GitHub Actions with Dockerized deployment support. Includes live webcam detection, model versioning, and monitoring dashboard.
- Tech: YOLOv8, ONNX, FastAPI, MLflow, Docker, GitHub Actions, React

### SafeSphere AI – AI-Powered Crime Prediction & Safety Platform

- DEVELOPED A SMART SAFETY PLATFORM THAT USES AI-BASED ANALYTICS AND GEOSPATIAL VISUALIZATION TO PREDICT CRIME-PRONE AREAS AND SUGGEST SAFER ROUTES. INCLUDES INTERACTIVE MAP INTERFACE, EMERGENCY SOS SYSTEM, REAL-TIME ALERTS, AND AI SAFETY ASSISTANT.
- TECH: REACT, TAILWIND, MAPBOX/GOOGLE MAPS API, AI APIS, FIREBASE

### MedVision AI – AI Medical Prescription Analyzer

- BUILT AN AI HEALTHCARE SYSTEM THAT EXTRACTS AND ANALYZES MEDICAL PRESCRIPTIONS USING OCR. PROVIDES MEDICINE EXPLANATIONS, DRUG INTERACTION WARNINGS, DOSAGE GUIDANCE, AND PATIENT-FRIENDLY SUMMARIES. INCLUDES SECURE AUTHENTICATION AND HISTORY TRACKING.
- TECH: REACT, FIREBASE, OCR (TESSERACT/GOOGLE VISION), AI API, TAILWIND CSS

### MedSecure Vault | *Python · Salsa20 · SHA-256 · Blockchain*

- Designed and built a tamper-proof medical prescription storage system using Salsa20 encryption and SHA-256 hashing with a blockchain-inspired architecture — ensuring integrity of sensitive patient data.

### Lawyer Case Management Dashboard | *Python · SQLite · PDF Generation*

- Developed a full-featured platform with role-based access control, case tracking & filtering, file handling, and automated PDF report generation — handling real-world legal workflow requirements.

### Library Management System | *Python · SQLite*

- Built an optimized system managing 500+ records with efficient CRUD operations and fast data retrieval — demonstrating strong database design and backend skills.

### Student Record System | *Python · GUI (Tkinter)*

- Created a GUI-based application for structured management, search, and updates of academic records with a clean, user-friendly interface.

### Inventory & Sales Dashboard | *Power BI*

- Designed interactive dashboards delivering actionable insights into sales trends and inventory levels — enabling data-driven business decisions.

### LexHub – UI/UX Prototype | *Figma*

- Produced a 50+ screen high-fidelity prototype applying modern UI principles, accessibility considerations, and complete user flow mapping for a legal tech platform.

## CERTIFICATIONS

---

- **Machine Learning** – [Simplilearn](#): Regression, classification, clustering, and model evaluation
- **Python Programming** – [Simplilearn](#): Core Python, data structures, and application development

## TECHNICAL SKILLS

---

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| Languages   | Python · Java · SQL                                                            |
| AI / ML     | Machine Learning · Regression · Classification · Clustering · Model Evaluation |
| Databases   | SQLite · MongoDB                                                               |
| Tools & BI  | Git · Power BI · Figma                                                         |
| Design      | UI/UX Design · Prototyping (50+ screen experience)                             |
| Soft Skills | Problem Solving · Attention to Detail · Self-Starter · Fast Learner            |

## WHAT I BRING TO YOUR TEAM

---

- Security-first mindset — built encryption systems from scratch, not just tutorials
- End-to-end project ownership — from database schema to user-facing GUI
- Data storytelling — translate raw data into Power BI dashboards stakeholders love
- Fast learner — self-taught ML, UI/UX, and blockchain concepts outside coursework
- Available immediately for full-time or part-time internship (remote or on-site in Punjab)